



schweizerischer ingenieur- und architektenverein
société suisse des ingénieurs et des architectes
società svizzera degli ingegneri e degli architetti
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Testspezifikation

SIA 4010

Test 7

Standort	Zürich Kloten		
Klima	SIA 2028 DRY normal, Zürich Kloten		
Simulationsperiode	1.1.2022 bis 31.12.2022		
Gebäude	Testgebäude gemäss Dokumentation		
Kälte-Verteilgruppen	Test 4	Luftkühler	
	Test 5	Luftkühler	
	Test 6	Luftkühler	
	Test 5 + 6	Kühldecken	
	Verbrauchsprofile	Es sind einheitliche Profile gemäss <i>Lastverläufe_220607.xlsx</i> zu verwenden	
Wärme-Verteilgruppen	Warmwasserladung	Ladeprofil (Wärme am Eingang des Warmwasserspeichers), gemäss <i>Lastverläufe_220607.xlsx</i>	
		Temperatur: 60°C	
	Test 4	Lufterwärmer	
	Test 5	Lufterwärmer	
	Test 6	Lufterwärmer	
	Test 5	Heizdecken	
	Test 6	Konvektoren	
	Verbrauchsprofile	Es sind einheitliche Profile gemäss <i>Lastverläufe_220607.xlsx</i> zu verwenden	
Kälteverteilung	Wärmeverluste	pauschal	5% der aufgenommenen Wärme
	Hilfsenergie	pauschal	2% der aufgenommenen Wärme davon als Wärmelast im Kälteverteilkreis: 50%
Wärmeverteilung	Wärmeverluste	pauschal	5% der abgegebenen Wärme
	Hilfsenergie	pauschal	2% der abgegebenen Wärme davon zurückgewonnen im Wärmeverteilkreis: 50%
Kältespeicherung	Wasserspeicher	Technischer Speicher zur Vermeidung kurzer Laufzeiten	
	Laderegulung	Volumen	2'000 l
		Ladung ein	Temperatur Speicher unten > Kältester Verteilkreis
Wärmespeicherung	Laderegulung	Ladung aus	Temperatur Speicher oben < Kältester Verteilkreis
		Ladung ein	Temperatur Speicher oben < Wärmster Verteilkreis
	Wasserspeicher	Technischer Speicher zur Vermeidung kurzer Laufzeiten	
	Laderegulung	Volumen	2'000 l
		Ladung aus	Temperatur Speicher unten > Wärmster Verteilkreis

Kälte- und Wärmeerzeugung

Typ

Leistung

Kennfeld Kälteerzeugung

Climaveneta NX-W-Y /H 0182

Wasser-Wasser-Wärmepumpe, reversibel, Scrollverdichter, 2-stufig

Hinweis: Kann als separate Kälte- und Wärmeerzeugung modelliert werden

Nennleistung Kälte: 55.9 kW

Nennleistung Wärme: 60.0 kW

NX-W /0182			
Outdoor side heat exchanger of chiller	air or water/brine		Water/brine
Indoor side heat exchanger chiller	water		Water
Type	compressor driven vapour compression or sorption process		Compressor driven vapour compression
Driver of compressor	electric motor or fuel driven, gaseous or liquid fuel, internal or external combustion engine		Electric motor
Rated cooling capacity	Prated,c	[kW]	55,9
Seasonal energy efficiency of the space cooling	ηs,c	[%]	208,0
Declared cooling capacity for part load at given outdoor temperatures Tj			
Declared cooling capacity at given outdoor temperatures Tj = 35°C	Pdc	[kW]	55,9
Declared cooling capacity at given outdoor temperatures Tj = 30°C	Pdc	[kW]	41,2
Declared cooling capacity at given outdoor temperatures Tj = 25°C	Pdc	[kW]	33,8
Declared cooling capacity at given outdoor temperatures Tj = 20°C	Pdc	[kW]	35,9
Degradation coefficient for chillers	Cdc		0,9
Declared energy efficiency ratio or gas utilisation efficiency/auxiliary energy factor for part load at given outdoor temperatures Tj			
Declared energy efficiency ratio at given outdoor temperatures Tj = 35°C	EERd	[%]	4,96
Declared energy efficiency ratio at given outdoor temperatures Tj = 30°C	EERd	[%]	5,76
Declared energy efficiency ratio at given outdoor temperatures Tj = 25°C	EERd	[%]	6,67
Declared energy efficiency ratio at given outdoor temperatures Tj = 20°C	EERd	[%]	7,17
Power consumption in modes other than "active mode"			
Off mode	POFF	[kW]	0,000
Thermostat-off mode	PTO	[kW]	0,476
Crankcase heater mode	PCK	[kW]	0,140
Standby mode	PSB	[kW]	0,023
Other items			
Capacity control	fixed/staged/variable		Staged
Sound power level, outdoor	LWA	[dB(A)]	74,0
GWP of the refrigerant		[Kg CO2eq]	2088
For air-to-water comfort chillers: air flow rate, outdoor measured		[m³/h]	-
For water/brine-to-water chillers: Rated brine or water flow rate, outdoor side heat exchanger		[m³/h]	11
Standard rating conditions used:	low temperature application/medium temperature application		Low temperature application

Testbedingungen EN 14825 (Zusatzinfo zu obiger Tabelle):

Tabelle 5 – Teillastbedingungen für Wasser/Sole-Wasser/Sole-Geräte und DX-Wasser-Geräte

Teillastverhältnis	Teillastverhältnis	Außenwärmeüberträger			DX	Innenwärmeüberträger			
		Kühlraum oder Wasser/Solekreislauf Eintritts-/Austrittstemperatur von Wasser/Sole	Erdegekoppelte Außenwärmeüberträger Eintritts-/Austrittstemperaturen von Wasser/Sole	Trockenkühler Eintritts-/Austrittstemperaturen des Wassers/Sole		Gebälsekonvektor Eintritts-/Austrittstemperaturen des Wassers/Sole		Kühldecken-Anwendung Eintritts-/Austrittstemperaturen des Wassers/Sole	
						Fester Austritt	Variabler Austritt ^c		
	%	°C	°C	°C	°C	°C	°C	°C	
A	(35-16)/(T _{design,c} - 16)	100	30/35	10/15	50/55	30	12/7	12/7	23/18
B	(30-16)/(T _{design,c} - 16)	74	26 ^b	10 ^b	45 ^b	30	*/7	*/8,5	*/18
C	(25-16)/(T _{design,c} - 16)	47	22 ^b	10 ^b	40 ^b	30	*/7	*/10	*/18
D	(20-16)/(T _{design,c} - 16)	21	18 ^b	10 ^b	35 ^b	30	*/7	*/11,5	*/18
^a Bei dem in der „A“-Prüfung bestimmten Volumenstrom für Geräte mit einem festen Volumenstrom oder bei einem festen ΔT = 5 K für Geräte mit einem variablen Volumenstrom. Falls für eine der Prüfbedingungen der resultierende Volumenstrom unterhalb des Mindestvolumenstroms liegt, wird dieser Mindestvolumenstrom als fester Volumenstrom mit der Austrittstemperatur für diese Prüfbedingung verwendet.									
^b Bei dem in der „A“-Prüfung bestimmten Volumenstrom für Geräte mit einem festen Volumenstrom oder bei einem festen ΔT = 5 K für Geräte mit einem variablen Volumenstrom. Falls für eine der Prüfbedingungen der resultierende Volumenstrom unterhalb des Mindestvolumenstroms liegt, wird dieser Mindestvolumenstrom als fester Volumenstrom mit der Eintrittstemperatur für diese Prüfbedingung verwendet.									
^c Liegt die variable Austrittstemperatur über dem Höchstwert des Betriebsbereichs des Geräts, dann sollte dieser Höchstwert betrachtet werden.									

Kennfeld
Wärmeer-
zeugung

NX-W-Y /H /0182			
LOW TEMPERATURE application			
Air-to-water heat pump:	yes / no		no
Water-to-water heat pump:	yes / no		yes
Brine-to-water heat pump:	yes / no		no
Low-temperature heat pump:	yes / no		no
With supplementary heater:	yes / no		no
Mixed unit with heat pump:	yes / no		no
Temperature application (1)	(low 35 °C/ medium 55 °C)		low 35 °C
Water flow rate	fixed / variable		fixed
Outlet temperature	fixed / variable		variable
Parameters are declared for average/warmer/colder climate conditions (1)	average / warmer / colder		average
Rated heat output at Tdesignh	Prated = Pdesignh	[kW]	76
Seasonal space heating energy efficiency	ηs	[%]	227
Seasonal space heating energy efficiency class	-	-	A+++
Declared capacity for heating for part load at indoor temperature 20 °C and outdoor temperature Tj			
Declared capacity for heating with outdoor temperature Tj = - 7 °C	Pdh	[kW]	66,8
Declared capacity for heating with outdoor temperature Tj = +2 °C	Pdh	[kW]	40,7
Declared capacity for heating with outdoor temperature Tj = +7 °C	Pdh	[kW]	36,1
Declared capacity for heating with outdoor temperature Tj = +12 °C	Pdh	[kW]	36,6
Declared capacity for heating with outdoor temperature Tj = Bivalent temperature	Pdh	[kW]	66,8
Declared capacity for heating with outdoor temperature Tj = Operation limit temperature	Pdh	[kW]	66,5
For air-to-water heat pumps: Tj = - 15 °C (if TOL < - 20 °C)	Pdh	[kW]	-
Bivalent temperature	Tbiv	[°C]	-7
Degradation coefficient	Cdh	-	0,90
Declared coefficient of performance or primary energy ratio for part load at indoor temperature 20 °C and outdoor temperature Tj			
Declared coefficient of performance with outdoor temperature Tj = - 7 °C	COPd	-	5,79
Declared coefficient of performance with outdoor temperature Tj = +2 °C	COPd	-	6,04
Declared coefficient of performance with outdoor temperature Tj = +7 °C	COPd	-	6,32
Declared coefficient of performance with outdoor temperature Tj = +12 °C	COPd	-	6,67
Declared coefficient of performance with outdoor temperature Tj = Bivalent temperature	COPd	-	5,79
Declared coefficient of performance with outdoor temperature Tj = Operation limit temperature	COPd	-	5,66
For air-to-water heat pumps: Tj = - 15 °C (if TOL < - 20 °C)	COPd	-	-
For air-to-water HP : Operation limit temperature	TOL	[°C]	-
Heating water operating limit temperature	WTOL	[°C]	60
Power consumption in modes other than active mode			
Off mode	POFF	[kW]	0,000
Thermostat-off mode	PTO	[kW]	0,943
Standby mode	PSB	[kW]	0,023
Crankcase heater mode	PCK	[kW]	0,140
Supplementary heater			
Nominal heating capacity	Psup	[kW]	9,02
Other items			
Capacity control	fixed / variable		variable
Sound power level, indoors	LWA	[dB(A)]	75
Sound power level, outdoors	LWA	[dB(A)]	-
Annual electricity consumption for heating	QHE	[kWh]	26600
Outdoor heat exchanger			
For air-to-water HP: Rated air flow rate, outdoors	Qairsorce	[m³/h]	-
For water-/brine-to-water heat pumps: Rated brine or water flow rate, outdoor heat exchanger	Qwater/brine source	[m³/h]	16
NX-W-Y /H /0182			
MEDIUM TEMPERATURE application			
Air-to-water heat pump:	yes / no		no
Water-to-water heat pump:	yes / no		yes
Brine-to-water heat pump:	yes / no		no
Low-temperature heat pump:	yes / no		no
With supplementary heater:	yes / no		no
Mixed unit with heat pump:	yes / no		no
Temperature application (1)	(low 35 °C/ medium 55 °C)		medium 55 °C
Water flow rate	fixed / variable		fixed
Outlet temperature	fixed / variable		variable
Parameters are declared for average/warmer/colder climate conditions (1)	average / warmer / colder		average
Rated heat output at Tdesignh	Prated = Pdesignh	[kW]	68
Seasonal space heating energy efficiency	ηs	[%]	181
Seasonal space heating energy efficiency class	-	-	A+++
Declared capacity for heating for part load at indoor temperature 20 °C and outdoor temperature Tj			
Declared capacity for heating with outdoor temperature Tj = - 7 °C	Pdh	[kW]	60,0
Declared capacity for heating with outdoor temperature Tj = +2 °C	Pdh	[kW]	36,5
Declared capacity for heating with outdoor temperature Tj = +7 °C	Pdh	[kW]	33,9
Declared capacity for heating with outdoor temperature Tj = +12 °C	Pdh	[kW]	34,9
Declared capacity for heating with outdoor temperature Tj = Bivalent temperature	Pdh	[kW]	60,0
Declared capacity for heating with outdoor temperature Tj = Operation limit temperature	Pdh	[kW]	59,1
For air-to-water heat pumps: Tj = - 15 °C (if TOL < - 20 °C)	Pdh	[kW]	-
Bivalent temperature	Tbiv	[°C]	-7
Degradation coefficient	Cdh	-	0,90
Declared coefficient of performance or primary energy ratio for part load at indoor temperature 20 °C and outdoor temperature Tj			
Declared coefficient of performance with outdoor temperature Tj = - 7 °C	COPd	-	3,82
Declared coefficient of performance with outdoor temperature Tj = +2 °C	COPd	-	4,83
Declared coefficient of performance with outdoor temperature Tj = +7 °C	COPd	-	5,54
Declared coefficient of performance with outdoor temperature Tj = +12 °C	COPd	-	6,33
Declared coefficient of performance with outdoor temperature Tj = Bivalent temperature	COPd	-	3,82
Declared coefficient of performance with outdoor temperature Tj = Operation limit temperature	COPd	-	3,52
For air-to-water heat pumps: Tj = - 15 °C (if TOL < - 20 °C)	COPd	-	-
For air-to-water HP : Operation limit temperature	TOL	[°C]	-
Heating water operating limit temperature	WTOL	[°C]	60
Power consumption in modes other than active mode			
Off mode	POFF	[kW]	0,000
Thermostat-off mode	PTO	[kW]	0,502
Standby mode	PSB	[kW]	0,023
Crankcase heater mode	PCK	[kW]	0,140
Supplementary heater			
Nominal heating capacity	Psup	[kW]	8,72
Other items			
Capacity control	fixed / variable		variable
Sound power level, indoors	LWA	[dB(A)]	75
Sound power level, outdoors	LWA	[dB(A)]	-
Annual electricity consumption for heating	QHE	[kWh]	29609
Outdoor heat exchanger			
For air-to-water HP: Rated air flow rate, outdoors	Qairsorce	[m³/h]	-
For water-/brine-to-water heat pumps: Rated brine or water flow rate, outdoor heat exchanger	Qwater/brine source	[m³/h]	13

